

Climate Impact Analysis of Bat Lake Salamanders (*A. maculatum*)

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This report describes the methods and results of a data-scientific pipeline forecasting survivorship probabilities of a northern population of yellow spotted salamanders under alternate climate scenarios. The writing and figures presented here form part of a larger manuscript on *A. maculatum* survivorship. All code and data used to complete this sub-analysis are viewable at <https://github.com/lepasserin/a-maculatum-climate-impact>.

1 Abstract

Temperate regions experience pronounced interannual climatic variability, providing the opportunity to understand the drivers of temporal variation in survival, and to model future impacts of environmental change. Temperate amphibians are sensitive to climatic variation and are experiencing dramatic declines in range size and abundance under global environmental change. We developed a priori hypotheses for climate effects on amphibian survival, then combine fifteen years of mark-recapture data from a population of spotted salamanders (*Ambystoma maculatum*) with Bayesian hierarchical modelling to test the association between environmental variables and annual survival. We found that greater temperatures during the spring inbound breeding migration led to a 50% decline in annual survival, while greater precipitation during the outbound migration was associated with a survival increase of 30%. Evaporative water loss may be the major driver of survival variation during the spring migration.

We then projected estimates of adult survival to the year 2100 under three climate change scenarios, incorporating the significant effects of inbound temperature and outbound precipitation. In the mildest scenario, projected increases in spring precipitation intensity were able to compensate for the negative effects of warmer spring temperatures, resulting in end-of-century survivorship estimates that were larger than current ones. However, faced with more extreme future climates, survivorship was projected to fall below current estimates. Our study draws urgent attention to the migratory dynamics of temperate, spring breeding amphibians faced with a changing climate.

2 Methods: Projecting Survivorship under Future Climate Change

We paired results from the posthoc breeding migration model with climate forecasts to estimate survival probabilities for the population of spotted salamanders in Bat Lake through the remainder of the 21st century. To generate these estimates, we first forecasted the inbound temperature and outbound precipitation covariates used in the breeding migration model using climate data from global circulation models (GCMs). Then, these forecasts were used to predict future survival probabilities of adults in the population.

Estimates of mean temperature, minimum temperature and total precipitation were required for each day of March, April, and May from 2005 to 2100 to provide reasonable estimates of the migration model’s environmental covariates. These estimates were sourced from global circulation model (GCM) output. In particular, we used the Coupled Model Intercomparison Project Phase 6 (CMIP6) dataset, which provides our required weather predictions for each day up until 2100 (Eyring et al. 2016). CMIP6 offers statistically downscaled weather predictions at spatial resolutions of 10 km², made publicly available by Environment and Climate Change Canada (Environment and Climate Change Canada 2024). We used the closest available grid point to the Algonquin East Gate Climate Station (45° 32’ N, 78° 17’ W), approximately 2.2 km away. For each climate variable considered, we used the ensemble mean of 26 different GCMs for our estimate. Estimates were recorded for three distinct climate

scenarios, representing conservative (SSP1), moderate (SSP2) and extreme (SSP5) fossil-fueled development respectively (see Riahi et al. 2017 for more detailed descriptions of these).

We applied two statistical bias correction algorithms to the GCM-derived climate forecasts, in accordance with standard practice for climate impact analysis (Sharma et al. 2007; Hagemann et al. 2011). For the mean and minimum temperature estimates, additive linear scaling was used to correct for long-run bias in the downscaled temperature data, following the procedure described in Hempel et al. (2013). For total precipitation, local intensity scaling was used to correct the intensity and frequency of rainy days, applied as described in Smitha et al. (2018). In both cases, the algorithms were first applied to GCM-derived estimates over a 20-year “historical” period (2005-2024), where they could be directly compared to the Algonquin East Gate Climate Station’s recorded observations. Once suitable correction factors had been identified, the correction algorithms were applied to future GCM-derived estimates data. Supplement A provides the explicit correction equations that were used, as well as visual comparisons between simulated, observed, and bias-corrected climate estimates.

Once bias-corrected, daily climate forecasts were used to predict the salamander breeding season start and end dates, for each year through to 2100. This allowed us to forecast for the specific environmental covariates used in the post-hoc breeding migration model, and, ultimately, annual survivorship. For a given year, the breeding season start date (equivalently, the Bat Lake ice out date) was predicted using that year’s mean April temperature, leveraging their strong, negative correlation observed during the BLISS study period, from 2008 to 2022 (Figure S2). The breeding season end date was always set to occur 22 days later, representing the average breeding season length observed during the same time period. Once determined, estimated breeding season intervals were used to calculate inbound temperature and outbound precipitation indices (see their equations in the section above). These indices were validated by comparing their hindcasted and observed values over the BLISS study period (Supplement B). Using the fitted breeding migration model, annual survivorship estimates were calculated for each year from 2005 to 2100, for each of the three climate scenarios described previously.

3 Results: Projecting Survivorship under Future Climate Change

Figure 1 displays estimates of annual adult survival for each year between 2008 and 2100, based on three future climate scenarios. The horizontal dashed line indicates adult survivorship estimated at the average environmental index values observed during the BLISS study period (2008 to 2022), and is referred to hereafter as the “baseline survival estimate” (~ 0.72). MA-smoothed estimates are emphasized in the plot to better distinguish the decadal trends produced by each scenario.

Barring the pronounced drop in average adult survival estimates from 2015 to 2030 in the most extreme case (SSP5), all three climate scenarios resulted in predictions that hovered around the baseline survivorship estimate until around 2060. Afterwards, the climate scenarios begin to produce divergent trends. Estimates derived from the climate scenarios SSP2 and SSP5 (representing moderate and extreme levels of fossil-fueled development respectively) both decrease below the baseline estimate, averaging around 0.66 in the final decade of the forecast. By contrast, survival in the most conservative scenario that we considered (SSP1) is expected to have increased relative to the baseline estimate, averaging around 0.77 in the last decade.

Looking at the raw trends (the more transparent lines in Figure 1), all climate scenarios produce survival estimates with pronounced interannual variability, especially past 2060. Before 2060, annual estimates range between 0.55 and 0.87, and between 0.45 and 0.91 afterwards. Further, annual survival estimates fall below 0.6 more often after 2060 in every scenario, but are rarer in earlier years.

Notably, the favourable survivorship estimates from SSP1 were not observed when only the temperature effect from the post-hoc breeding migration model was considered. Further, estimates from SSP5 decreased dramatically, approaching annual estimates of around 0.5 by the end of the century see Supplement B. These observations suggest that the significant effect of outbound precipitation detected by the post-hoc breeding migration model has an important compensatory effect on survival for the spotted salamanders of Bat Lake.

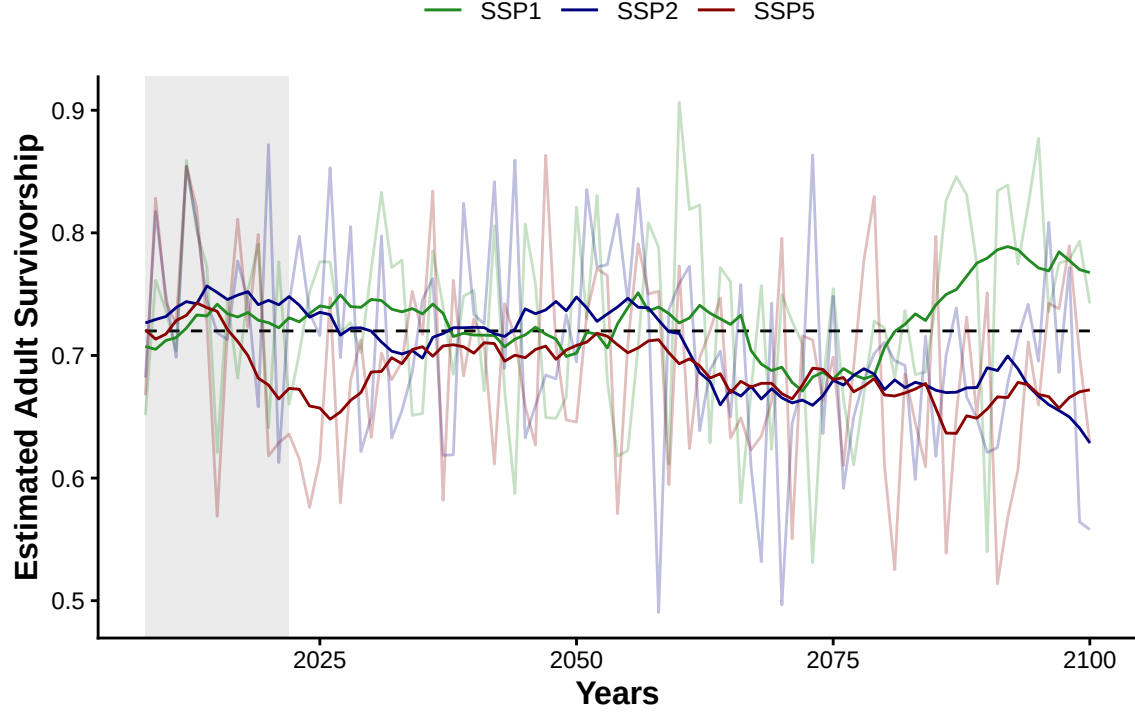


Figure 1: Time series of estimated adult survivorship from 2008 to 2100, based on environmental data derived from global climate model projections. Red-, green-, and blue-colored trendlines each represent different climate scenarios (see main text for details). Bolded lines show projections after MA smoothing was applied ($q = 6$), and raw projections are shown as transparent lines. The grey-shaded band indicates the BLISS study period (2008 to 2022), and the horizontal dashed lines indicate the adult survivorship estimated at the average environmental values observed during this period.

Supplement

A Statistical Bias Corrections

The daily projections of mean temperature, minimum temperature, and total precipitation used in this study (hereafter referred to as “climate projections”) were sourced originally from the Coupled Model Intercomparison Project Phase 6 (CMIP6) dataset, which provides daily estimates of these climate variables under various socio-political scenarios (called “Shared Socioeconomic Pathways”, often abbreviated as SSPs). Since GCM output is coarse-grained, Environment and Climate Change Canada provides downscaled estimates with higher spatial resolution [10 km²; Eyring et al. (2016)], for each of three SSPs (SSP1, SSP2, and SSP5 ; see (Riahi et al. 2017) for detailed definitions). These are more appropriate for the analysis of highly localized populations such as the Bat Lake spotted salamander population. Even with downscaling, however, climate projections are often still systematically biased (Sharma et al. 2007; Hagemann et al. 2011); while observed trends are often accurately reproduced, estimates may deviate substantially from observations in their absolute values and/or interannual variations.

We implemented two bias correction methods to correct specific, systematic deviations between observed and simulated climate metrics. In both cases, these methods were first applied over a historical period (2005 to 2024) where observed daily climate data was available near our study site (measurements taken from Algonquin East Gate Climate Station), to determine suitable correction factors. Then, these correction factors were applied to correct bias in future projections (through to 2100), under the assumption that systematic bias in the past will manifest itself in the same way in the future. We note here that projections from all three SSPs considered were corrected using the same coefficients.

A.1 Correcting Temperature Forecasts

For both mean and minimum daily temperature, downscaled estimates showed similar trends to observed measurements over the entire historical period, but with consistently higher monthly averages. Simulated temperatures also exhibited smaller day-to-day variation compared to observed trends, likely because our simulated estimates are based on the ensemble mean of 26 different GCMs, which artificially deflates variance. Methods that correct the variance and the mean in simulated temperature data simultaneously exist, but have been shown to impact the climate change signal in certain datasets (Hagemann et al. 2011). Given how temperature projections are used in further analyses, we opted to only correct the systematic bias in the average, preserving the climate change signal most faithfully. To this end we applied additive linear scaling following Hempel et al. (2013), implemented as described below.

Let i represent the March of a given year between 2005 and 2024. Then, let T_{ij}^{OBS} represent the observed temperature (mean or minimum) of the j^{th} day in that month respectively. T_{ij}^{SIM} and T_{ij}^{ADJ} are defined similarly, for simulated (SIM) and bias-corrected (ADJ) daily values. We calculate the linear scaling constant C as follows:

$$C = \left(\sum_{i=2005}^{2024} T_i^{\text{OBS}} - T_i^{\text{SIM}} \right) / 20 \quad (1)$$

where T_i^{OBS} and T_i^{SIM} represent the average temperature of the i^{th} March from 2005 to 2024, for observed and simulated data respectively. C can then be used to scale future daily temperature simulations like so:

$$T_{i,j}^{\text{ADJ}} = C + T_{i,j}^{\text{SIM}} \quad (2)$$

Notice that C is calculated over the entire period, and not per-year. This allows us to apply the correction shown in Equation 2 to every March in the forecasted simulated data. This procedure can then be repeated for April and May, mean and minimum temperatures separately,

resulting in the estimation of 6 linear scaling constants to correct GCM-derived temperature forecasts.

A.2 Correcting Precipitation Forecasts

For daily total precipitation, downscaled estimates had similar monthly and seasonal averages as observed measurements, but grossly underestimated the day-to-day variance in precipitation, effectively predicting a drizzle each day. This resulted in an underestimated number of rainy days, and an underestimated intensity of wet days. Since the precipitation main effect used in our analysis is sensitive to the number of rainy days (see main text), this drizzle bias required correction. To this end we used local intensity scaling following Smitha et al. (2018), described below.

Let i represent the March of a given year between 2005 and 2024. Then, let P_{ij}^{OBS} represent the observed precipitation (total) on the j^{th} day in that month respectively. P_{ij}^{SIM} and P_{ij}^{ADJ} are defined similarly, for simulated (SIM) and bias-corrected (ADJ) daily values. First, a fixed threshold t is determined such that the average number of wet days with simulated precipitation totals below t is equal to the average number of observed dry days. This constant is used to correct for the number of dry days in the simulated data. Then, we calculate the observed average precipitation among wet days (D_1), and the average precipitation among simulated days above t (D_2). Their ratio, $D = \frac{D_1}{D_2}$, is used to correct for the expected intensity of wet days in the simulated data. These constants are then applied to the simulated precipitation data as follows:

$$P_{i,j}^{\text{ADJ}} = \begin{cases} 0, & \text{if } P_{i,j}^{\text{SIM}} < t \\ (P_{i,j}^{\text{SIM}} - t) \times D, & \text{otherwise} \end{cases} \quad (3)$$

Similarly to temperature, three sets of (D, t) pairs were fit, for each month in our dataset.

A.3 Bias Correction Results & Diagnostics

All fitted correction factors are shown in Table A1. Monthly biases in simulated temperatures were more extreme earlier in the season, and for minimum vs. average daily temperature estimates. Simulated precipitation days were deemed dry when their total precipitation did not exceed around 2mm, equivalent to a light shower. Simulated wet days saw their total precipitation amount inflated by a factor of 4 to 7. By construction, the biases we targeted for each climate variable were completely removed when values are averaged over the entire period. However, we also found that, for any specific month in the historical data, raw GCM data was more likely to deviate from observed estimates than bias-corrected GCM data.

Figure A1 compares observed, simulated, and bias corrected daily weather estimates for an arbitrarily chosen year in the historical period (2010). This is shown to illustrate how the correction algorithms modify the simulated data; the effects were qualitatively similar for any chosen year within the historical period (2005 to 2024). Since all temperature bias correction coefficients are negative, the simulated trend is pulled downwards. For precipitation, days receiving under ~2mm of precipitation are deemed dry, and wet days see their precipitation estimates inflated, creating daily variability that is similar to the observed trend.

Table A1: Fitted correction factors applied to simulated climate projections.

Correction Factor	Value	Correction Factor	Value
March mean temperature offset (°C)	-1.54834	March dry day threshold (mm)	2.11449
April mean temperature offset (°C)	-1.38055	April dry day threshold (mm)	2.16254
May mean temperature offset (°C)	-0.57552	May dry day threshold (mm)	2.57951
March minimum temperature offset (°C)	-2.91662	March intensity coefficient (unitless)	4.03366
April minimum temperature offset (°C)	-1.90356	April intensity coefficient (unitless)	5.97328
May minimum temperature offset (°C)	-1.66153	May intensity coefficient (unitless)	6.87419

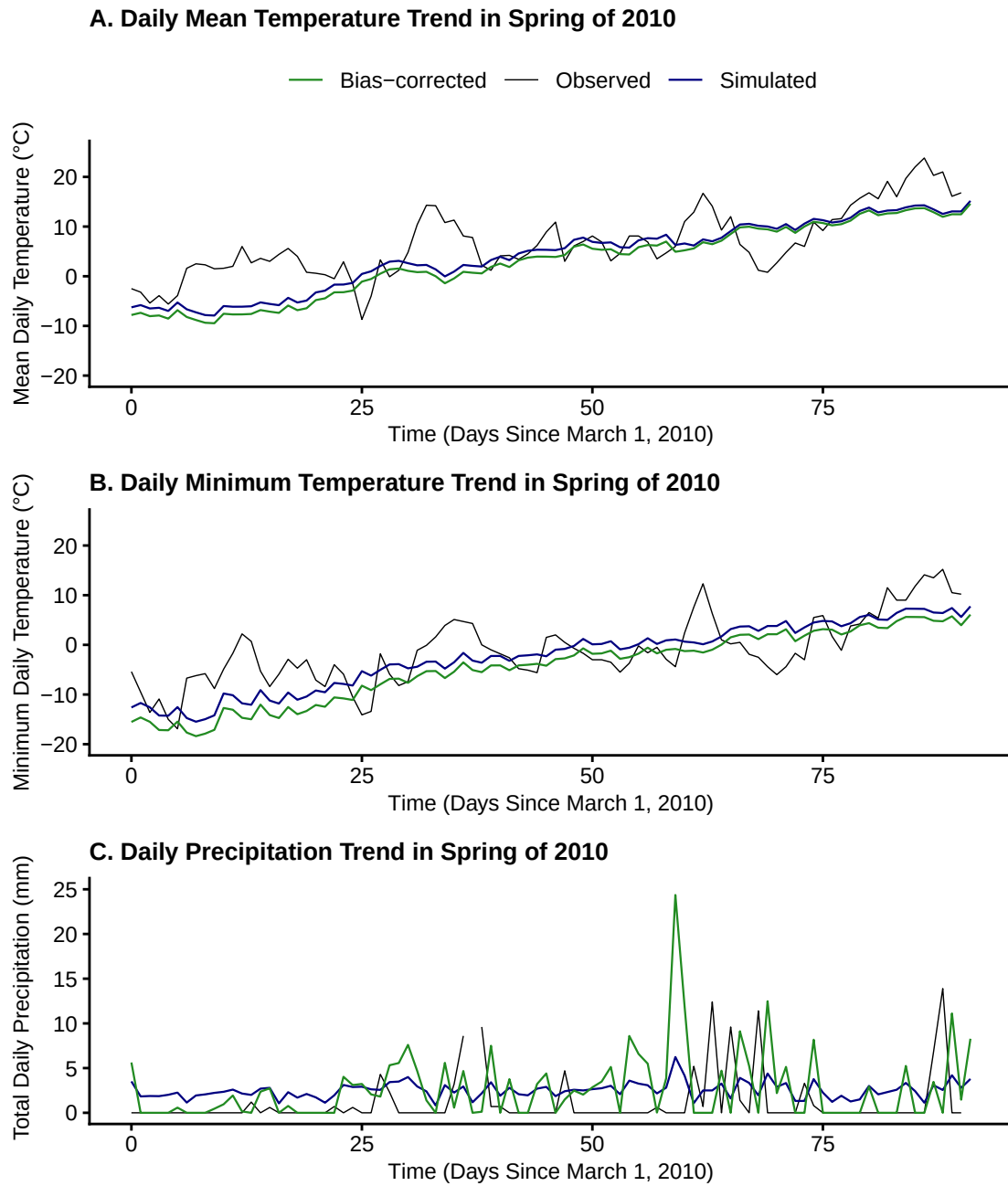


Figure A1: Time series of observed (black), simulated (blue), and bias-corrected (green) daily weather estimates from March 1 to May 31, 2010. Panel A displays mean daily temperature, panel B displays minimum daily temperature, and panel C displays total daily precipitation.

B Forecasting Environmental Indices & Survival Probabilities

Bias-corrected daily weather forecasts were used to calculate four key measures, for each year from 2005 to 2100. First, average mean daily temperatures in April were used to predict the breeding season start date (during the BLISS study period, these measures were strongly correlated; $|\rho| = 0.73$). Breeding season end date was calculated directly from the start date, by adding the average breeding season length observed during the BLISS study period. Using the forecasted breeding season interval, we calculated the inbound temperature and outbound precipitation indices as they were calculated in the posthoc breeding migration model (see main text for their formulas). This series of predictions and calculations was made for each climate scenario separately.

Figure B1 displays the resulting time series of estimates for the breeding season start date and both climate indices. These graphs are also overlaid with the observed values during the BLISS study period. Average index hindcasts over this period were very similar to observed averages, for all climate scenarios. We note, however, that observed outbound temperature indices displayed more annual variability than hindcasted ones.

Forecasted climate indices (panel B and C in Figure B1) were used to estimate adult survival probabilities over the century, using the posthoc breeding migration model fit in the main text. The survival probability at year t ($S_a(t)$) was derived from the temperature (Temp) and precipitation (Precip) indices estimated for that year using the following formula:

$$S_a(t) = \text{logit}^{-1} [\beta_0 + \beta_1 \text{Temp}(t) + \beta_2 \text{Precip}(t)] \quad (4)$$

where β_1 , β_2 , and β_3 were fit using the breeding migration model. The resulting survival forecast for each climate scenario is shown in the Results section of the main text.

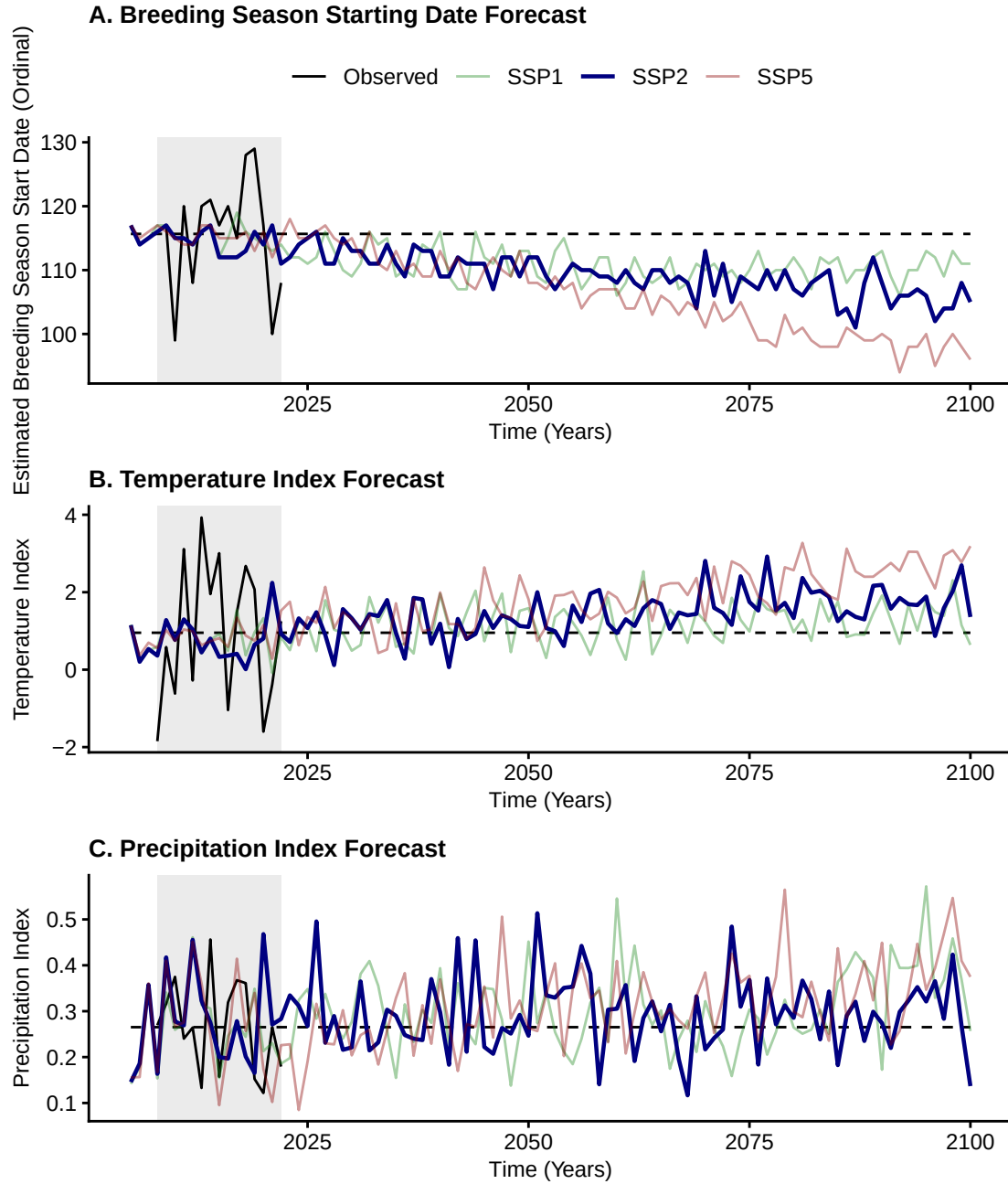


Figure B1: Time series of observed (black), and forecasted (coloured) indices, from 2005 to 2100. Panels A, B, and C display breeding season start date, inbound temperature, and outbound precipitation indices respectively. The grey shaded areas represent the BLISS study period (2008 to 2022), and the horizontal dashed lines indicate the average observed index values during this period. The climate scenario labelled ‘SSP2’ is emphasized for visual contrast.

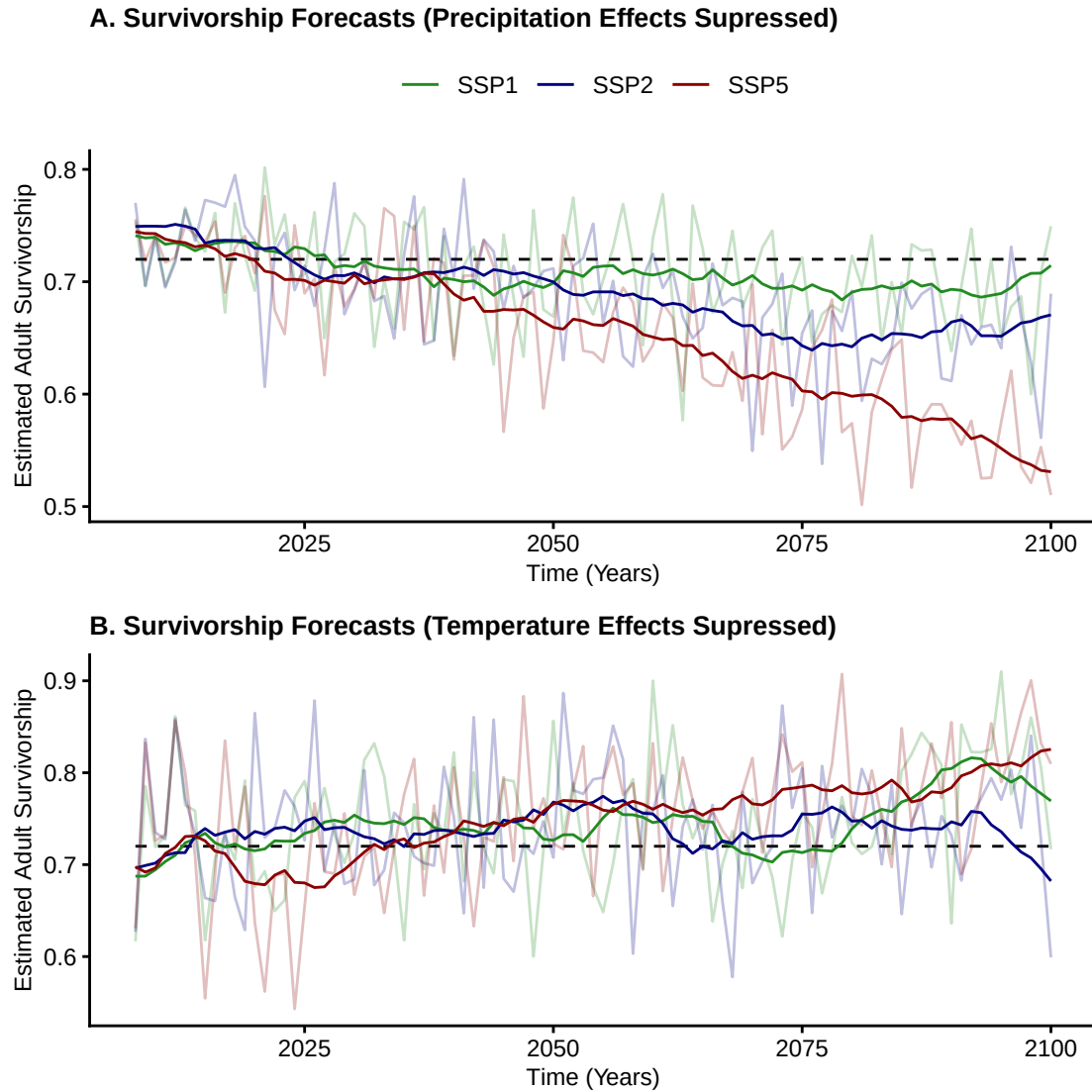


Figure B1: Effect-suppressed forecasts of adult survivorship of Bat Lake spotted salamanders, until 2100. Red, green, and blue forecasts each represent three different climate scenarios (see main text for details). Bolded lines show projections after MA smoothing was applied ($q = 6$), and raw projections are shown as transparent lines. Smoothed trends are emphasized to better distinguish the climate scenarios. (A) displays the forecast when only the effect of spring temperature is included, and (B) displays the forecast when only the effect of outbound precipitation is included. The horizontal dashed lines indicate the adult survivorship estimated at the average environmental values observed during the BLISS study period (2008 to 2022).

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